[D.3.2] Universal AI: AIXI

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Iliad Intensive

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The big picture: one agent, two halves

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- **The common engine:** a single **Bayesian mixture** ξ over a class $\mathcal{M}$ of candidate environments, weighted by a prior w_ν.
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- **Universal** choice: $\mathcal{M} =$ all computable environments, prior $w_\nu = 2^{-K(\nu)}$ (K is the "complexity" of the environment). Assumption "$\mu \in \mathcal{M}$" becomes "*the universe is computable*".

The Bayesian mixture ξ

Definition

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- **Key fact (mixture is a predictor):** its one-step prediction is a *posterior-weighted* average of the members' predictions (you prove this today!)

$$\xi(x_t \mid x_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \, \nu(x_t \mid x_{<t}), \qquad w(\nu \mid x_{<t}) = w_\nu \, \frac{\nu(x_{<t})}{\xi(x_{<t})}.$$

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- **Posterior update is multiplicative:** $w(\nu \mid x_{1:t}) = w(\nu \mid x_{<t}) \cdot \dfrac{\nu(x_t \mid x_{<t})}{\xi(x_t \mid x_{<t})}$: we reweight by how well ν predicted vs. ξ (Bayes' rule!)

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$$h = a_1\, e_1\, a_2\, e_2\, a_3\, e_3\, \cdots \qquad \text{(history } æ_{<t}\text{)}$$

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- Spaces: actions $\mathcal{A}$, observations $\mathcal{O}$, rewards $\mathcal{R} \subset [0,1]$, percepts $\mathcal{E} = \mathcal{O} \times \mathcal{R}$, and (finite) histories $\mathcal{H}^* := (\mathcal{A} \times \mathcal{E})^* := \cup_{t=0}^{\infty} (\mathcal{A} \times \mathcal{E})^t$.

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- **No Markov / MDP assumption!** Both the policy and environment can depend on all prior percepts.

The three players

Policy π

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Interaction measure ν^π

Run π inside ν and multiply along the history:

$$\nu^\pi(æ_{t:m} \mid æ_{<t}) := \prod_{k=t}^{m} \pi(a_k \mid æ_{<k})\, \nu(e_k \mid æ_{<k} a_k).$$

Value, optimality, and AIXI

Value function (assuming geometric discount γ)

$$V_\nu^{\pi,m}(æ_{<t}) = (1-\gamma)\,\mathbb{E}_\nu^\pi[G_{t-1:m} \mid æ_{<t}] \text{ with } G_{t-1:m} = \sum_{k=t}^{m} \gamma^{k-t}\, r_k$$

$$= (1-\gamma) \sum_{æ_{t:m}} \nu^\pi(æ_{t:m} \mid æ_{<t}) \sum_{k=t}^{m} \gamma^{k-t}\, r_k$$

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- **Optimal policy** π_ν^* satisfies $V_\nu^{\pi_\nu^*} = V_\nu^*$.
- **Bayes-optimal policy** $\pi_\xi^* \in \arg\max_\pi V_\xi^\pi$: acts optimally w.r.t. the *mixture* ξ.

AIXI

π^*_ξ with $\mathcal{M}$ = all ≈computable environments[a] and prior $w_\nu = 2^{-K(\nu)}$. Write ξ_U for ξ with this choice of $\mathcal{M}$ and w_ν, so AIXI $= \pi^*_{\xi_U}$.

[a] lower-semicomputable semi-measures

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AIXI, explicitly

max over actions, $\sum$ over percepts, choosing actions that maximise the ξ-weighted discounted return:

$$a_t = \arg\max_{a_t} \lim_{m\to\infty} \sum_{e_t} \max_{a_{t+1}} \sum_{e_{t+1}} \cdots \max_{a_m} \sum_{e_m} \xi(e_{t:m} \mid æ_{<t}\, a_{t:m}) \sum_{k=t}^{m} \gamma^{k-t} r_k$$

Result 1: On-policy value convergence

Theorem

For any $\mu \in \mathcal{M}$ and any policy π,

$$V_\xi^\pi(æ_{<t}) - V_\mu^\pi(æ_{<t}) \to 0 \qquad \mu^\pi\text{-almost surely.}$$

Intuition: *Following policy π, pretending the environment is ξ, eventually will act as if we knew the true environment μ.*

Result 2: AIXI can't be fooled

Theorem (deterministic μ)

If $V^*_\mu(æ_{<t}) > \varepsilon$ along history $æ_{<t}$, then

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$$V_\xi^*(æ_{<t}) \;\geq\; w_\mu\,\varepsilon \;>\; 0.$$

Intuition: *Whenever good behaviour in μ is achievable, the Bayes-optimal agent secures at least a fraction proportional to the prior weight of μ.*

Result 3: Self-optimizing property (stretch goal)

Theorem

Let $\mathcal{M}$ be finite and fix a historic policy π generating the actions $a_{<t}$. If *some* policy can learn to act optimally in every $\nu \in \mathcal{M}$, then π^*_ξ does too: for all $\mu \in \mathcal{M}$,

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- **But not for AIXI:** the theorem needs $\mathcal{M}$ finite. Over all computable environments AIXI explores too little and is *not* asymptotically optimal [Ors13, LH15], though some other policies are (see the safety slides).

Takeaways

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- **The catch:** both are *incomputable* (K is incomputable; $\mathcal{M}$ is intractable to enumerate, the expectimax is intractable). They are gold standards, not computable algorithms, and the prior's $2^{-K(\nu)}$ hides a dependency on the choice of universal Turing machine: a bad choice of UTM can make AIXI behave arbitrarily badly [LH15].

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- **Approximations:** MC-AIXI-CTW [VNH+11] gives an approximation using Context Tree Weighting to replace ξ and Monte-Carlo-Tree-Search to replace the expectimax.

Why care for safety? (I) What would an optimal agent do?

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- **Reward hijacking.** Given a delusion box, a reward maximizer uses it [RO11]. An agent planning in an unknown world, unable to rule out that *the reward signal itself* is what it should optimize, is incentivized to intervene in the provision of reward [CHO22]; this argument underpins a *Science* policy paper on long-term planning agents [CKB+24].

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- **Exploration vs. traps.** Asymptotic optimality is achievable (e.g. Thompson sampling [LLOH16]), but only by trying everything: an asymptotically optimal agent is eventually destroyed or incapacitated [CCH20].

Why care for safety? (II) Mitigations and the bridge to LLMs

- **Myopia + boxing.** BoMAI maximizes reward only within an episode, in a sealed room; it is *asymptotically unambitious*, i.e. indifferent to seizing outside power [CVH20].

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- **Embedded agency.** AIXI sits outside its world and can't model itself [WH25]. Reflective oracles give a grain of truth for multi-agent play [WHLT25] and self-predicting universal agents [M$^+$25].

Want to work on this? AIXI Labs

- **AIXI Labs** ("AI X-risk Initiative"): an AI safety org founded in 2026 by Cole Wyeth, Aram Ebtekar, Michael Cohen, Matthias Dellago and Marcus Hutter [WEC+26].

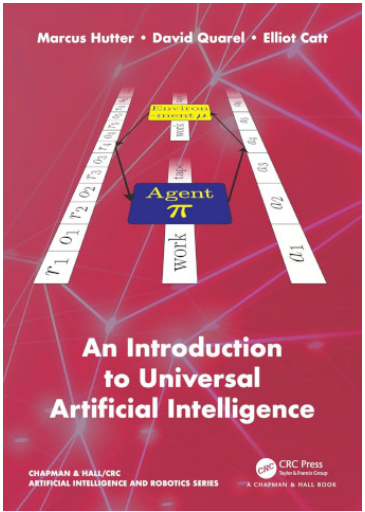
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- **Focus:** algorithmic information theory, continual RL, and AIXI. Stated agenda:
  - strengthen the *technical* case that ASI poses an existential risk;
  - prototype mitigations built on conservative AIXI variants (pessimism, corrigibility) and push frontier labs to adopt them;
  - port the AIXI analysis to LLM agents and test it empirically;
  - long-term deconfusion: learning in unrealizable environments, corrigibility.

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- Funded by UK AISI's Alignment Project; runs a research fellowship and is hiring. For the worldview, read "The AIXI perspective on AI Safety" [Wye26].

Further reading: the textbook



An Introduction to Universal Artificial Intelligence
Hutter, Quarel & Catt, 2024 [HQC24]

- Today's material follows the book.
- Gentle intro, late-undergrad level.
- Covers prediction, agents, approximations and safety.
- Free PDF and lecture slides with exercises.

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